

Name: \_\_\_\_\_

Class: \_\_\_\_\_

**SYDNEY TECHNICAL HIGH SCHOOL****YEAR 12 ASSESSMENT TASK****EXTENSION 2****MATHEMATICS****MARCH 2006****Instructions**

- Attempt all questions
- Answers to be written on the paper provided
- Start each question on a new page
- Marks may not be awarded for careless or badly arranged working
- Indicated marks are a guide and may be changed slightly if necessary
- These questions must be handed in attached to the top of your solutions.

|     |     |     |       |
|-----|-----|-----|-------|
| Q1  | Q2  | Q3  | Total |
| /16 | /17 | /17 |       |

**Question 1 (16 marks)**

- a) If  $z = -1 + i$  find, (4)
- (i)  $\bar{z}$
- (ii)  $|z|$
- (iii)  $\arg(iz)$
- b) i. Express  $z = -\sqrt{3} - i$  in modulus – argument form (3)
- ii. Hence write  $z^{12}$  in the form  $x + iy$ , where  $x$  and  $y$  are real
- c) Find all complex numbers  $z$ , such that  $z^3 = 64i$  (3)

- d) Given that  $a$  and  $b$  are real numbers, find  $a$  and  $b$  if,  $\frac{5+2i}{a+bi} = 1+i$  (3)
- e) On an argand diagram shade the region containing all points representing complex numbers  $z$  such that  $2 \leq \operatorname{Re}(z) \leq 5$  and  $-2 \leq \operatorname{Im}(z) \leq 4$

**Question 2** (17 marks) (Start a new page)

- a) Evaluate  $\int_0^1 x\sqrt{1-x} \, dx$  using a suitable substitution. (3)

- b) Consider the ellipse  $3x^2 + 4y^2 = 12$  (6)

- Determine the eccentricity of the ellipse
- Find the coordinates of the foci  $S$  and  $S'$  and also the equation of the directrices.
- Sketch the ellipse showing all important information.

c)

- Show that the equation of the tangent to the ellipse  $\frac{x^2}{9} + \frac{y^2}{4} = 1$  at the point

$P(3\cos\theta, 2\sin\theta)$  is given by  $\frac{x\cos\theta}{3} + \frac{y\sin\theta}{2} = 1$  (3)

The ellipse above cuts the  $y$  axis at the points  $A$  and  $B$  (5)

The tangents to the ellipse at  $A$  and  $B$  meet the tangent to the ellipse at  $P(3\cos\theta, 2\sin\theta)$  at the points  $C$  and  $D$  respectively.

- Draw a neat diagram showing the positions of  $P, A, B, C$  and  $D$ .
- Show that  $AC \times BD = 9$

Start a new page

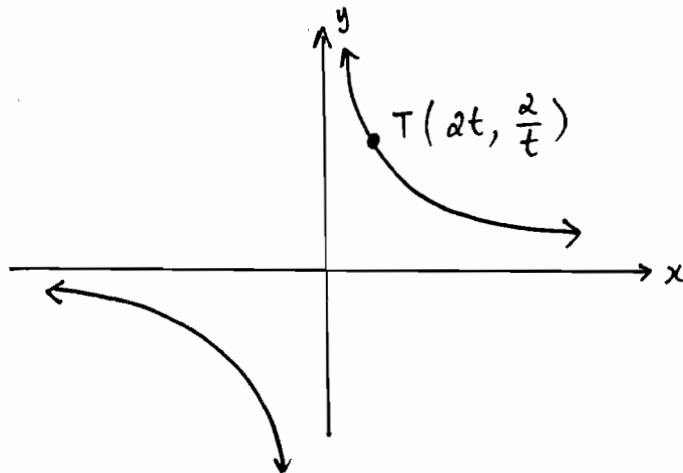
**Question 3** (17 marks)

- a) Show that the locus specified by  $3|z - (4 + 4i)| = |z - (12 + 12i)|$  is a circle, (4)

Write down its radius and the coordinates of its centre

- b) Find the equation of the tangent to the curve  $x^2 + xy^2 - 6y = 0$  at the point (2,1) (4)

- c) Consider the diagram below



- i. Show that the tangent to the hyperbola  $xy = 4$  at the point  $T(2t, \frac{2}{t})$  has equation  $x + t^2y = 4t$ . (2)
- ii. This tangent cuts the  $x$ -axis at point Q. Show that the line through Q which is perpendicular to the tangent at T has equation  $t^2x - y = 4t^3$ . (2)
- iii. This line through Q cuts the rectangular hyperbola at the points R and S. Show that the midpoint M of RS has coordinates  $M(2t, -2t^3)$ . (2)
- iv. Find the equation of the locus of M as T moves on the rectangular hyperbola, stating any restrictions that may apply. (3)

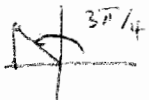
**End of Test**

## Solutions

### Question 1

a)  $\bar{z} = -1 - i$  ✓

$\sqrt{2}$  ✓

$\arg(z) = \frac{3\pi}{4}$  

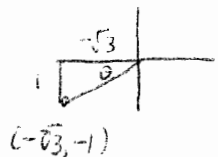
$\arg(iz) = \pi/2 + 3\pi/4$

$= 5\pi/4$  ✓

$= -\frac{3\pi}{4}$  ✓

b)  $z = -\sqrt{3} - i$

$\angle = -\frac{5\pi}{6}$



$|z| = 2$

$\therefore z = 2 \operatorname{cis} \left( -\frac{5\pi}{6} \right)$  ✓

ii.  $z^{12} = 2^{12} \operatorname{cis} \left( 12 \times -\frac{5\pi}{6} \right)$  ✓

$= 4096 (1 + i0)$

$= 4096$

✓ ie  $2^{12}$

c)  $z^3 = 64i$

$z^3 - 64i = 0$

$z^3 + (4i)^3 = 0$

$(z + 4i)(z^2 - 4iz - 16) = 0$

$z = -4i$  ✓,  $z = 2i \pm 2\sqrt{3}$  ✓

d)  $5 + 2i = (1+i)(a+bi)$  ✓

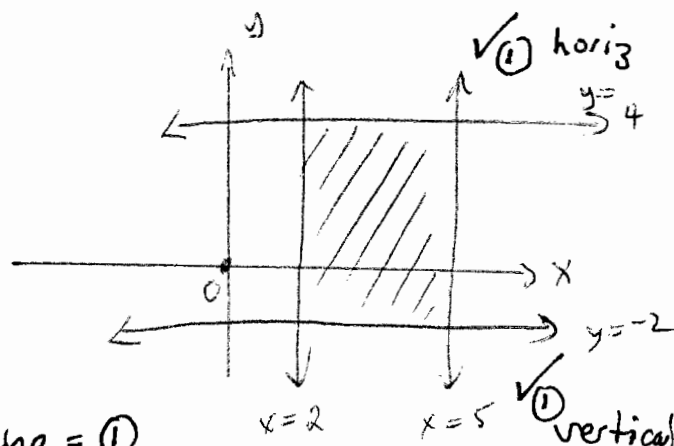
$= a + bi + ai - b$

$\therefore a - b = 5$

$a + b = 2$  ✓

$2a = 7$

e)



shading = ①

### Question 2

a)  $u = 1 - x$

$x = 1 - u$

$\frac{du}{dx} = -1$

$x = 1 \quad u = 0$

$x = 0 \quad u = 1$

$-du = dx$

①

$\therefore \int_1^0 (1-u)\sqrt{u} \cdot -du$  ①

$= \int_1^0 \sqrt{u} - u^{3/2} du$

$= \left[ \frac{2}{3} u^{3/2} - \frac{2}{5} u^{5/2} \right]_0^1$

$= \left( \frac{2}{3} - \frac{2}{5} \right) - (0)$

$= \frac{4}{15}$

①

$$b) \frac{x^2}{4} + \frac{y^2}{3} = 1 \quad \checkmark$$

$$a = 2 \quad b = \sqrt{3}$$

$$i. \quad b^2 = a^2(1 - e^2)$$

$$\frac{3}{4} = 1 - e^2$$

$$e^2 = \frac{1}{4} \quad \checkmark$$

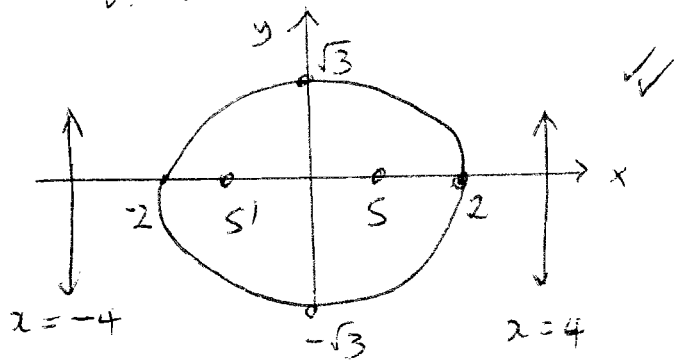
$$e = \frac{1}{2}$$

$$S = (ae, 0) \quad S'(-ae, 0)$$

$$S = (1, 0) \quad S'(-1, 0) \quad \checkmark$$

$$\text{directrices } x = \pm a/e$$

$$\therefore x = \pm 4 \quad \checkmark$$



$$c) \frac{2x}{9} + \frac{2y}{4} \cdot \frac{dy}{dx} = 0 \quad \checkmark$$

$$\frac{dy}{dx} = -\frac{2x}{9} \times \frac{4}{2y}$$

$$= -\frac{4}{9} \frac{x}{y}$$

$$M_T = \frac{-4 \cdot 3 \cos \theta}{3 \cdot 9 \cdot 2 \sin \theta}$$

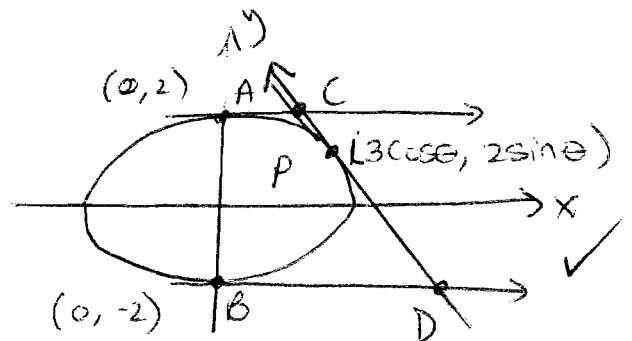
$$= \frac{-2 \cos \theta}{3 \sin \theta} \quad \checkmark$$

$$y - 2 \sin \theta = \frac{-2 \cos \theta}{3 \sin \theta} (x - 3 \cos \theta)$$

$$3 \sin \theta y - 6 \sin^2 \theta = -2 \cos^2 \theta + 6 \cos^2 \theta$$

$$2 \cos^2 \theta + 3 \sin \theta y = 6$$

$$\therefore \frac{\cos^2 \theta}{3} + \frac{\sin \theta y}{2} = 1$$



$$\text{tangent at A } y = 2$$

$$\text{at B } y = -2$$

$$\text{tangent at P}$$

$$\frac{x \cos \theta}{3} + \frac{y \sin \theta}{2} = 1 \quad (1)$$

$$\text{Finding C sub } y = 2 \text{ into (1)}$$

$$\frac{x \cos \theta}{3} + \sin \theta = 1$$

$$\frac{x \cos \theta}{3} = 1 - \sin \theta$$

$$\checkmark \quad x = \frac{3(1 - \sin \theta)}{\cos \theta}$$

$$C = \left[ \frac{3(1 - \sin \theta)}{\cos \theta}, 2 \right]$$

$$\text{Finding D sub } y = -2 \text{ into (1)}$$

$$\checkmark \quad \frac{x \cos \theta}{3} - \sin \theta = 1$$

$$D = \left[ \frac{3(1 + \sin \theta)}{\cos \theta}, -2 \right]$$

$$\begin{aligned}
 KC \times BD &= \frac{3(1-\sin\theta)}{\cos\theta} \times \frac{3(1+\sin\theta)}{\cos\theta} \\
 &= 9 \frac{(1-\sin^2\theta)}{\cos^2\theta} \\
 &= 9
 \end{aligned}$$

### Question 3

$$\begin{aligned}
 i) \quad 3|x+iy-4-4i| &= |x+iy-12-12i| \\
 3|(x-4) + (y-4)i| &= |(x-12) + (y-12)i| \\
 9[(x-4)^2 + (y-4)^2] &= (x-12)^2 + (y-12)^2
 \end{aligned}$$

$$\begin{aligned}
 9[x^2-8x+16+y^2-8y+16] &= x^2-24x+144 \\
 &\quad + y^2-24y+144
 \end{aligned}$$

$$8x^2 - 48x + 8y^2 - 48y = 0$$

$$x^2 - 6x + y^2 - 6y = 0$$

$$(x-3)^2 + (y-3)^2 = 18$$

$$\therefore \text{centre } (3,3) \quad \text{radius} = 3\sqrt{2}.$$

b) Implicit diff

$$2x + 1 \cdot y^2 + x \cdot 2y \cdot \frac{dy}{dx} - 6 \frac{dy}{dx} = 0 \quad (1)$$

$$2x + y^2 + \frac{dy}{dx}(2xy-6) = 0$$

$$\frac{dy}{dx} = \frac{-2x-y^2}{2xy-6} \quad (1)$$

$$\therefore \text{at } (2,1) \quad M_T = \frac{-2(2)-1}{2(2)(1)-6}$$

$$= \frac{5}{2} \quad (1)$$

$\therefore$  eq tangent

$$u-1 = 5/2(x-2) \quad \leftarrow (1)$$

$$c) \quad y = 4/x \quad y' = -\frac{4}{x^2} \quad \text{at } x=2t$$

$$\begin{aligned}
 i. \quad M_T &= -\frac{4}{4t^2} \\
 &= -1/t^2 \quad (1)
 \end{aligned}$$

$$\therefore \text{eq } y - \frac{2}{t} = -\frac{1}{t^2}(x-2t) \quad (1)$$

$$t^2y - 2t = -x + 2t$$

$$\therefore x + t^2y = 4t \quad \text{QED}$$

$$ii. \quad x \text{-int } y=0 \quad x=4t \quad \therefore Q(4t, 0)$$

$$M_{\perp} = t^2 \quad (1)$$

$$\begin{aligned}
 \text{eq: } y-0 &= t^2(x-4t) \\
 y &= t^2x - 4t^3
 \end{aligned}$$

$$\therefore t^2x - y = 4t^3$$

$$\begin{aligned}
 iii. \quad \text{Solve } t^2x - y &= 4t^3 \text{ and} \\
 xy &= 4 \text{ simultaneously}
 \end{aligned}$$

$$y = t^2x - 4t^3 \quad \& \quad xy = 4$$

$$\therefore x(t^2x - 4t^3) = 4$$

$$t^2x^2 - 4t^3x - 4 = 0 \quad (1)$$

as the 2 x values for this give R & S let them be  $\alpha$  &  $\beta$

$$\alpha + \beta = \frac{4t^3}{t^2} = 4t$$

$$\therefore \text{midpt } \frac{\alpha + \beta}{2} = 2t \quad (1)$$

$$\text{and } y \text{ value} = -2t^3$$

iv. Locus of m

$$x = 2t \rightarrow \frac{x}{2} = t$$

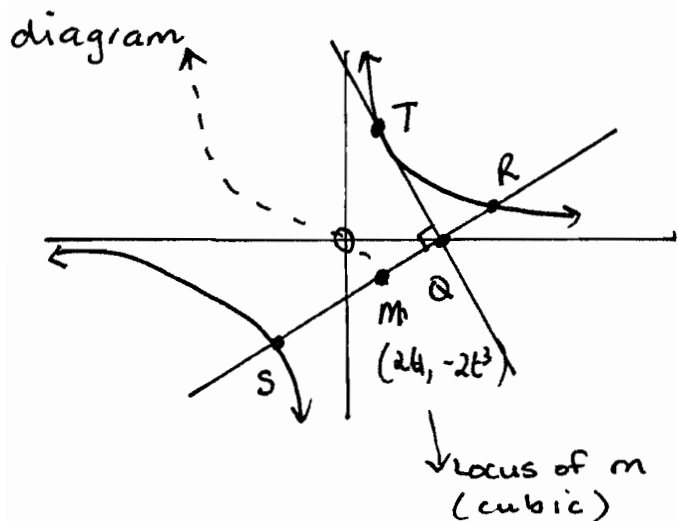
$$\text{Sub into } y = -2t^3$$

$$y = -2 \left( \frac{x}{2} \right)^3$$

$$y = -2 \cdot \frac{x^3}{8}$$

$$y = \frac{x^3}{-4}$$

$$-4y = x^3 \quad \textcircled{1}$$



Locus of m moves on the ①

cubic equation in the  
2nd & 4th quadrants ①

Restriction → excluding the  
origin as  $t \neq 0$ .

Alternative Sol<sup>n</sup> to Q3 (c) iii

$$t^2 x^2 - 4t^3 x - 4 = 0$$

$$x = \frac{4t^3 \pm 4t\sqrt{t^4 + 1}}{2t^2}$$

$$= 2t \pm \frac{2\sqrt{t^4 + 1}}{t}$$

$$R \left[ 2t + \frac{2\sqrt{t^4 + 1}}{t}, -2t^3 + 2t\sqrt{t^4 + 1} \right]$$

$$S \left[ 2t - \frac{2\sqrt{t^4 + 1}}{t}, -2t^3 - 2t\sqrt{t^4 + 1} \right]$$

$$\text{Midpt } M = \left[ \frac{4t}{2}, -\frac{4t^3}{2} \right]$$

$$= (2t, -2t^3)$$